

> whoami

Habtamu Asefa

AI Research Engineer / AI Products / Speech Models

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// summary

AI Research Engineer and founder building **PrepX** (2,600+ downloads, 93,060+ questions answered) and low-resource speech models at **Liqawnt**. Ships LLM agents, RAG, mobile apps, and backend services to production. Author of **react-native-ajora** (6,100+ npm downloads); previously founded **Temaribet**. BSc Biomedical Engineering, Addis Ababa University.

technical_skills

– 8 domains

AI engineering	LLM agents, RAG, tool calling, prompt engineering, evals, streaming AI UX, model routing (OpenRouter)	Search & retrieval	Qdrant (vector), Typesense (full-text), hybrid search, textbook & curriculum indexing
Backend	NestJS, FastAPI, Node.js, Express, REST APIs, auth, notification pipelines	Mobile	React Native, Expo, EAS Build, offline-first architecture, SSE streaming interfaces
Agent tooling	Agno, Mastra, AG-UI / CopilotKit patterns, server-side tool execution	Databases	PostgreSQL, MongoDB, SQLite
DevOps	AWS EC2, Debian Linux, Nginx, PM2, Certbot, SSL/DNS, production debugging	Languages	TypeScript, JavaScript, Python, SQL

experience

– git log --author=habtamu

2024 – Present

Addis Ababa, ET

React Native	NestJS
FastAPI	Qdrant
Typesense	RAG
OpenRouter	

Founder & Lead Engineer

PrepX – AI-native learning platform

Textbook-grounded AI tutoring with offline-first mobile UX for Ethiopian high-school students.

- Built and shipped the full **React Native (Expo)** app and AI tutoring pipeline; grew to **2,600+ downloads and 93,060+ questions answered**, and launched Free, Go, and Pro subscriptions with referral and leaderboard rewards.
- Designed a **hybrid RAG** system (**Qdrant** vector + **Typesense** full-text) for answers grounded in the curriculum, kept **model-agnostic** across Gemini, GPT, Claude, and open models via OpenRouter.
- Modeled a **self-referential curriculum graph** across Grades 9–12 — **1,800+ topics and 220,000+ practice questions**, with notes, quizzes, leveled worksheets, flashcards, and past entrance exams.
- Built backend services with NestJS, FastAPI, and PostgreSQL powering chat, retrieval, content workflows, notifications, and live competitions.

2019 – 2025

Addis Ababa, ET

Marketplace
Operations EdTech

Founder

Temaribet – tutoring marketplace

In-home tutoring marketplace connecting families with vetted college-student tutors.

- Built and operated a two-sided marketplace to **4,000+ tutors and 500+ families** (started as a freshman), owning onboarding, matching, operations, and product strategy.
- Selected into **Reach for Change** and **Orbit Innovation Hub** accelerators (backed by the IKEA Foundation & Mastercard Foundation); awarded grant funding and won a startup pitch competition.

2023 – Present

Remote · US / EU

React Native

Node.js

NestJS

MongoDB

AWS EC2

Nginx

Full-Stack / Mobile Engineer

Freelance & Agency – US / EU clients

Production mobile and web applications for international startups and agency teams.

- **Dymnd (Chicago startup):** shipped production React Native features used by **100+ users** over a multi-month paid engagement — rebuilt timezone-sensitive habit logic and fixed OTP, consent, keyboard, and navigation defects to improve reliability.
- Maintained and upgraded a **US bar-exam prep platform** — backend/infra work (Node.js, AWS EC2, Nginx, Certbot) plus production bug fixes and stability improvements.
- Owned delivery across both, start to finish: **EAS** builds, app-store releases, server deployment, and live production debugging.

selected_projects

– ls ./projects

2024 – Present

React Native

SSE

Gemini

npm

react-native-ajora — open-source AI agent UI for React Native

- Authored "the most complete AI agent UI for React Native" — **6,100+ npm downloads, 19 GitHub stars**. Ports AG-UI / CopilotKit agentic patterns to mobile: **SSE** streaming, tool calls, sources, threads, and server-side tool execution with Gemini.

Ongoing

TypeScript

React Native

Python

Open-source contributions

- Contributor to **Expensify's** production React Native app; **49 public repos** spanning AI, mobile, and backend.

2024

Embedded

Sensors

Biomedical

UlcerGuard — diabetic foot-ulcer prevention footwear

AAU / AAiT thesis

- Built an embedded pressure-sensing footwear prototype detecting high-pressure foot regions to help prevent diabetic foot ulcers — biomedical analysis, sensors, and hardware prototyping.

education

BSc, Biomedical Engineering

Addis Ababa University / AAiT · 2024

Thesis: **UlcerGuard** — pressure-sensing footwear for diabetic foot-ulcer prevention.

achievements

- § Scaled **Temaribet** to 4,000+ tutors & 500+ families; selected into **Reach for Change & Orbit Innovation Hub**.
- § Shipped **PrepX** to 2,600+ downloads and 93,060+ questions answered with a 3-person team.
- § Authored **react-native-ajora** — 6,100+ npm downloads, 19★.
- § Delivered production software for US/EU startups across mobile, backend & infra.

target_roles

> AI Research Engineer · Speech & Multimodal AI · Inference · Fine-tuning · Model Deployment